INCH-POUND
MIL-DTL-16377/84A
w/AMENDMENT 1
25 July 2022
SUPERSEDING
MIL-DTL-16377/84A
30 June 2014

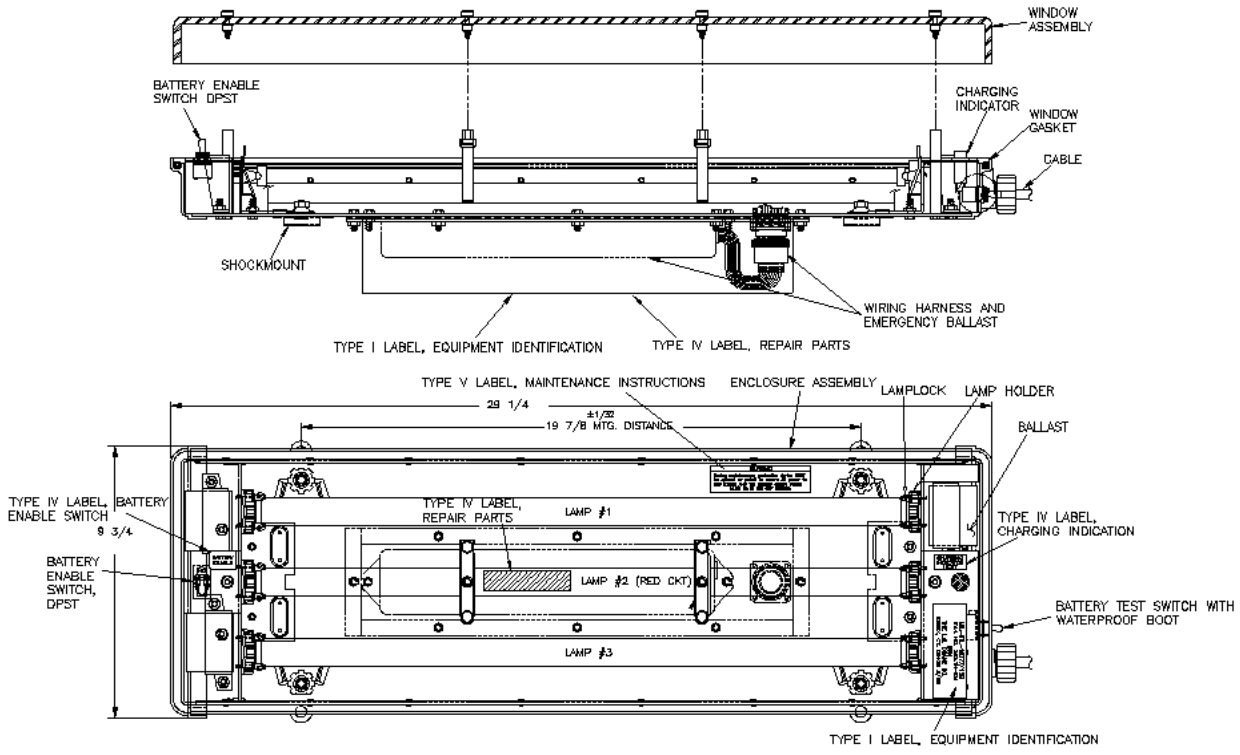
DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T8 FLUORESCENT, EMERGENCY LIGHTING, 51 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 333.1T8EM, 333.2T8EM, 342.2T8EM, AND 342.3T8EM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377. If specified by the acquisition activity, solid state lighting (SSL) fixtures designed and manufactured in accordance with MIL-DTL-16377/75 may be substituted for fixtures manufactured in accordance with this specification sheet.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.



NOTE:

1. Symbols 333.2T8EM and 342.3T8EM use identical construction with flush mounting flange not shown.

FIGURE 1. Typical view of symbols 333.1T8EM and 342.2T8EM.

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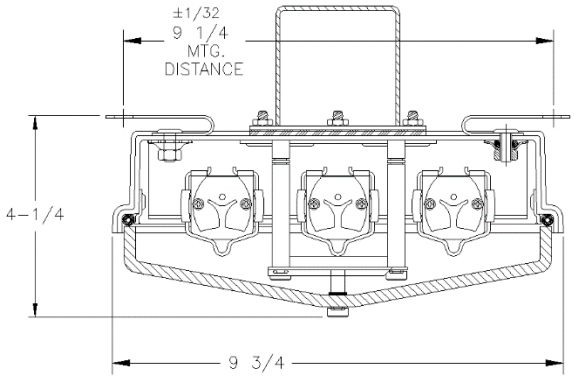
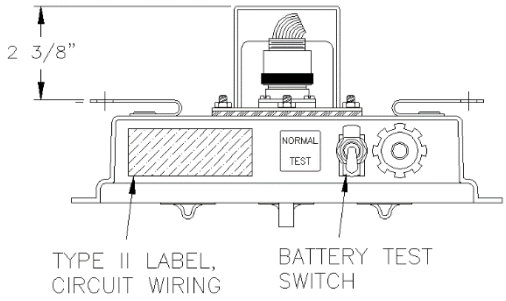


FIGURE 2. Typical section for symbols 333.1T8EM and 342.2T8EM.



NOTES:

1. Symbols 333.2T8EM and 342.3T8EM use identical battery test switch construction with additional switch hardware (not shown) for switch operation below ceiling.
2. Type II label required on all fixtures.

FIGURE 3. Typical section for symbols 333.2T8EM and 342.3T8EM.

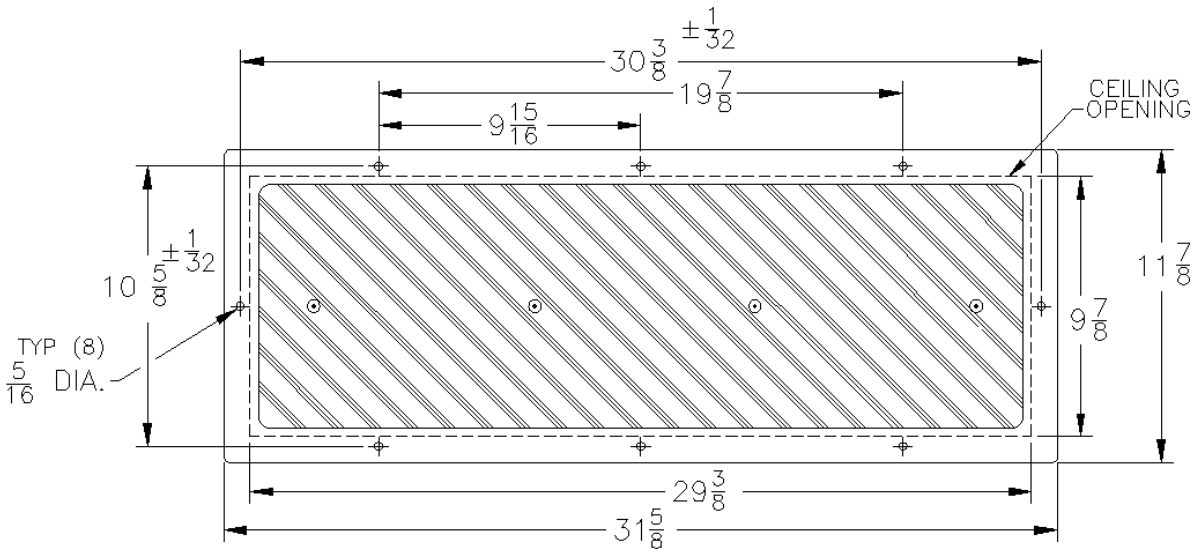


FIGURE 4. Flush mounting dimension for symbols 333.2T8EM and 342.3T8EM.

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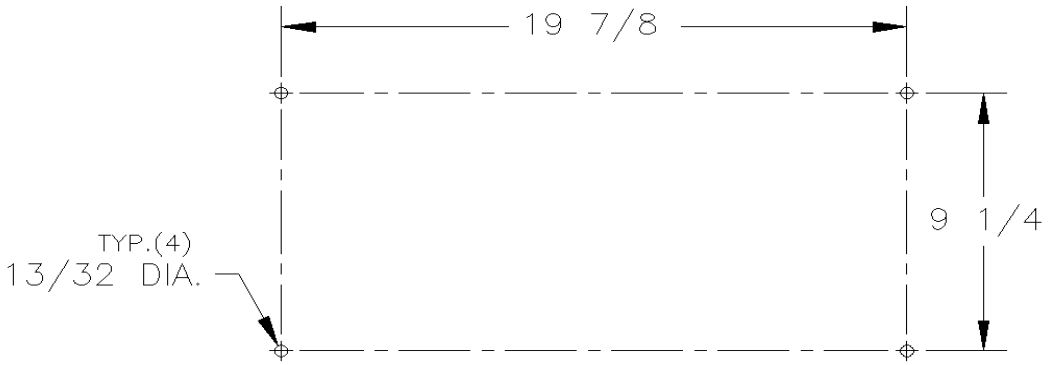
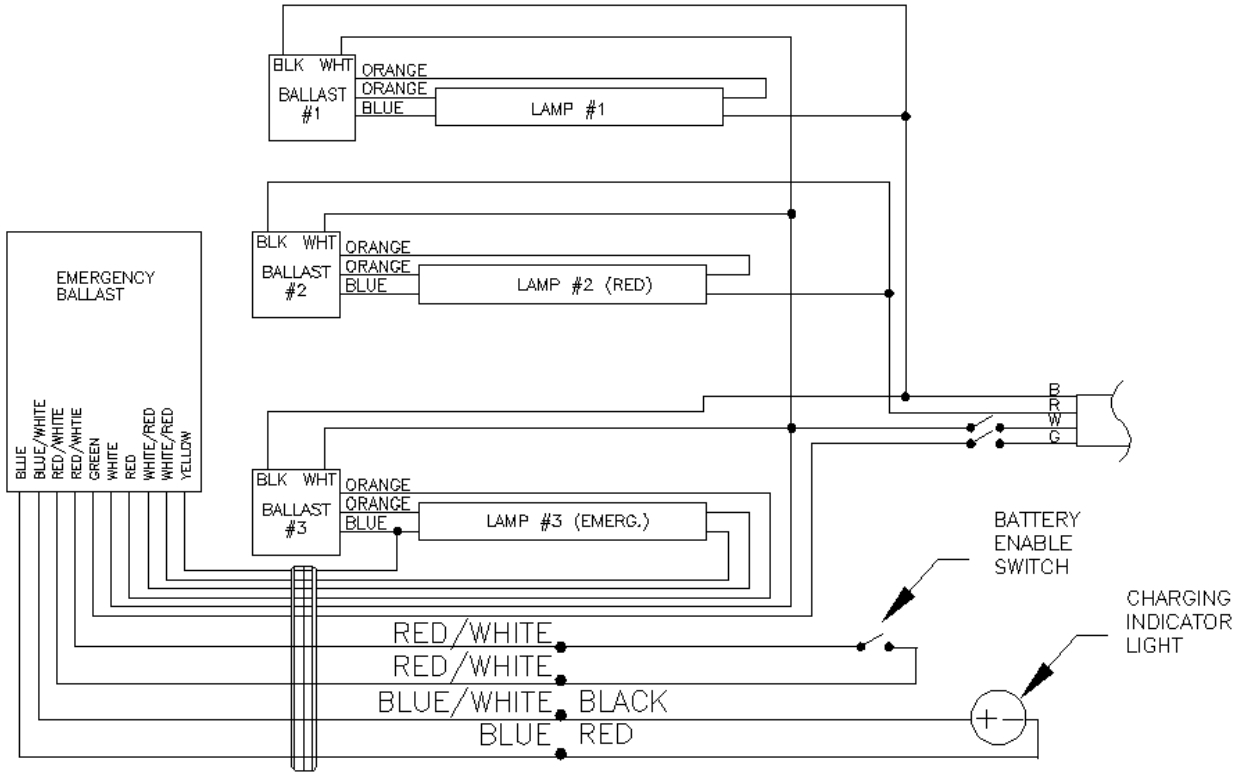


FIGURE 5. Standard mounting dimensions utilizing shock mounts for symbols 333.1T8EM and 342.2T8EM.



NOTES:

1. The black and red input conductors are the only switched power leads to the fixture, used for normal on/off lamp control.
2. The white and green input conductors shall be connected directly to the power source and remain continuously energized for emergency lamp battery charging. The white and green input conductors shall not have power interrupted by on/off control switches.
3. The white input conductor is shared by the normal lamp circuits (white and red) and by the emergency circuit.

FIGURE 6. Typical wiring diagram.

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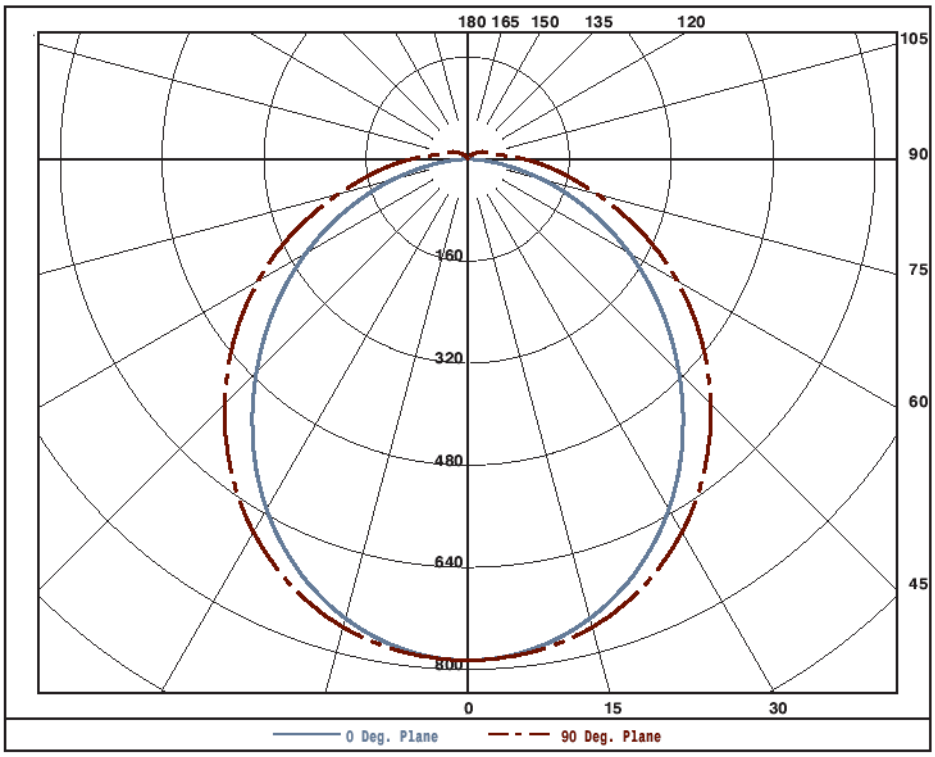


FIGURE 7. Candela distribution curves, white diffusing window, symbols 333.1T8EM and 333.2T8EM.

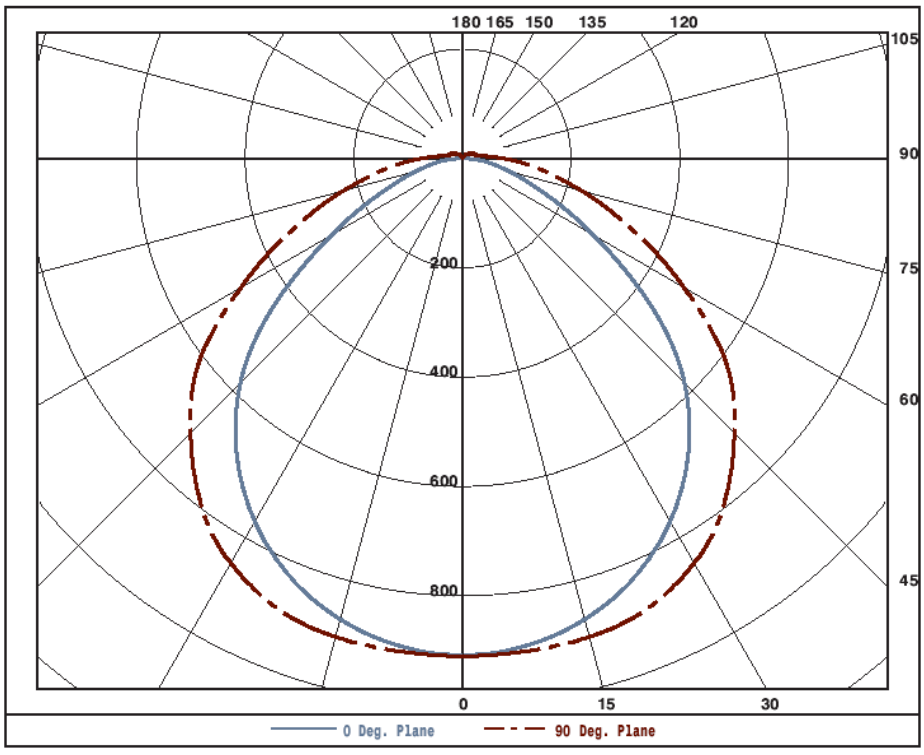


FIGURE 8. Candela distribution curves, prismatic window, symbols 342.2T8EM and 342.3T8EM.

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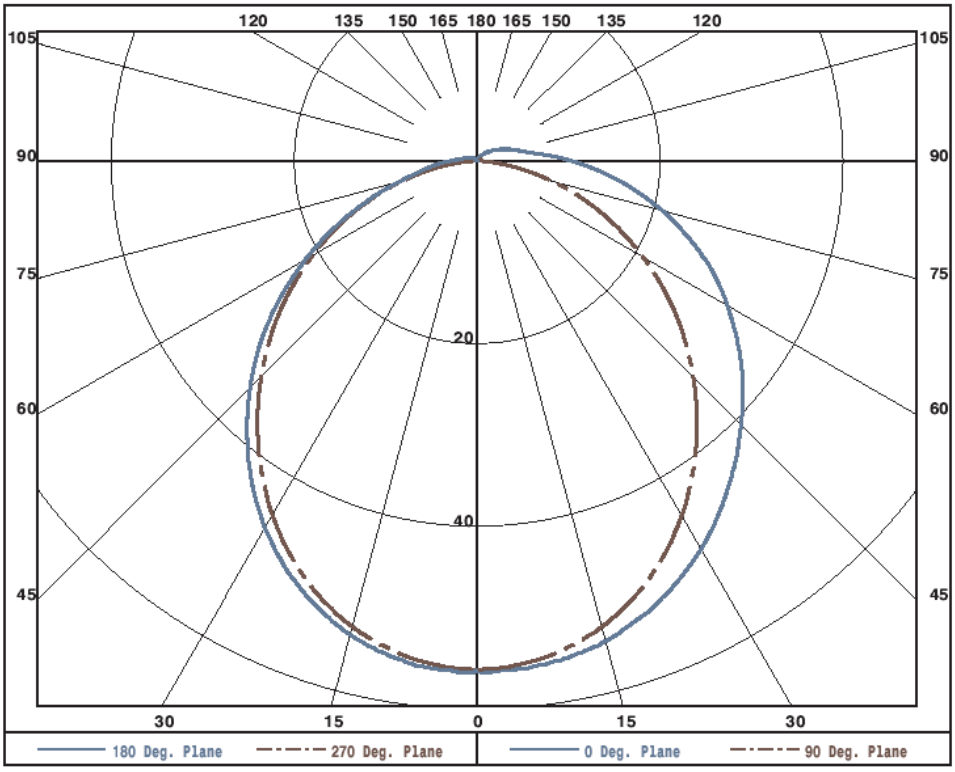


FIGURE 9. Candela distribution curves, emergency mode.

TABLE I. Numerical candela distribution, white diffusing window, symbols 333.1T8EM and 333.2T8EM.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	786	786
5	781	783
15	745	760
25	677	709
35	582	634
45	469	540
55	353	436
65	234	323
75	120	211
85	27	124
90	9	89

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	8	62
105	6	39
115	4	26
125	3	17
135	0	10
145	0	3
155	0	0
165	0	0
175	0	0
180	0	0

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TABLE II. Numerical candela distribution, prismatic window, symbol 342.2T8EM and 342.3T8EM.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	910	910
5	904	910
15	872	906
25	810	881
35	713	817
45	577	709
55	378	566
65	195	381
75	88	227
85	30	119
90	9	76

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	8	48
105	6	26
115	5	22
125	3	15
135	1	9
145	0	4
155	0	1
165	0	0
175	0	0
180	0	0

TABLE III. Numerical candela distribution, emergency mode.

Angle	Candela Distribution	
	0-Degree	90-Degree
0	56	56
5	56	55
15	54	53
25	51	48
35	46	41
45	41	33
55	35	25
65	28	17
75	21	8
85	14	2
90	10	1

Angle	Candela Distribution	
	0-Degree	90-Degree
95	8	0
105	4	0
115	3	0
125	2	0
135	1	0
145	1	0
155	0	0
165	0	0
175	0	0
180	0	0

PHOTOMETRIC DATA:

Photometry reports based on LM-41 and other pertinent Illuminating Engineering Society (IES) procedures.

Plane:	0 degrees is parallel to lamps, 90 degrees is perpendicular to lamps.
Electrical values:	120 volts, 60 hertz, 53.7 watts + 4.0 watts (battery charging circuit) = 57.7 watts total.
Ambient temperature:	77±1.8 °F (25±1 °C).
Lamps:	GE F17T8-SP41-ECO, or equal (seasoned for at least 100 hours).
Lamps correction:	Lamps rated at 1,325 lumens.
Ballast:	Furnished with fixture, see MIL-DTL-16377/81 for 17-watt lamp specification, MIL-DTL-16377/77 for emergency ballast specification.

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REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

333.1T8EM:	Complete lighting fixture assembly, standard mounting with three 17-watt cool white fluorescent lamps and white diffusing window assembly; includes emergency battery/ballast circuit for emergency lighting.
333.2T8EM:	Complete lighting fixture assembly, flush mounting with three 17-watt cool white fluorescent lamps and white diffusing window assembly; includes emergency battery/ballast circuit for emergency lighting.
342.2T8EM:	Complete lighting fixture assembly, standard mounting with three 17-watt cool white fluorescent lamps and clear and white prismatic window assembly; includes emergency battery/ballast circuit for emergency lighting.
342.3T8EM:	Complete lighting fixture assembly, flush mounting with three 17-watt cool white fluorescent lamps and clear and white prismatic window assembly; includes emergency battery/ballast circuit for emergency lighting.

DESIGN:

Electrical rating:	51 watts, 120 volts, 60 hertz.
Dimension and configurations:	See figures 1, 2, and 3.
Photometric data:	See figures 7, 8, and 9 and tables I, II, and III.
Wiring diagram:	See figure 6 for typical wiring.
Ballast:	See MIL-DTL-16377/81 for 17-watt lamp specification and MIL-DTL-16377/77 for emergency ballast specification.
Mounting dimensions:	
Symbols 333.1T8EM and 342.2T8EM:	Standard mounting (surface), see figure 5.
Symbols 333.2T8EM and 342.3T8EM:	Flush mounting (recessed), see figure 4.
Window assembly:	
Symbols 333.1T8EM and 333.2T8EM:	White window, see table IV and figure 27 of MIL-DTL-16377.
Symbols 342.2T8EM and 342.3T8EM:	Prismatic window, see table IV and figure 27 of MIL-DTL-16377.
Enclosure:	Watertight, see 3.5.11 of MIL-DTL-16377.
Cable:	10 feet (minimum) of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 333.1T8EM and 342.2T8EM:	Not to exceed 20 pounds.
Symbols 333.2T8EM and 342.3T8EM:	Not to exceed 22 pounds.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 1 and 3:

a. Two Type I labels, equipment identification, reference MIL-DTL-16377 figure 11; shall include manufacturer and address, one located on back and other located on an inner nonreflecting surface.

b. One Type II label, circuit wiring information, reference MIL-DTL-16377 figure 12; located on outside surface adjacent to cable entrance.

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c. Two Type IV labels, repair parts, reference MIL-DTL-16377 figure 14; information to be shown shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table IV, located on an inner nonreflecting surface and back of fixture.

d. One Type IV label, reference MIL-DTL-16377 figure 14; information to be shown shall be "CHARGING INDICATOR LIGHT."

e. One Type IV label, reference MIL-DTL-16377 figure 14; information to be shown shall be "BATTERY ENABLE."

f. One Type V label, reference MIL-DTL-16377 figure 15; information to be shown shall be "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION."

EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 as follows:

First article testing/inspection:
- Complete fixtures: See table V.
- Window assemblies: See table VI.

Conformance:
- Comparison inspection: See tables V and VI.
- Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is show in table VII.

NOTES:

1. See MIL-DTL-16377/42 for T8 color filter tube specifications and part numbers.
2. Installation circuit wiring shall incorporate a minimum ratio of eight 17-watt lamps and one emergency 17-watt lamp. The 8-to-1 ratio is required for meeting MIL-STD-1399-300-1 harmonic requirements. The 17-watt emergency ballast produces non-linear current that shall be offset by the linear current from eight 17-watt lamp ballasts connected to the same circuit.
3. Emergency fixtures may be designated as Navy Emergency Automatic Lighting System (NEALS).
4. The "battery enable" switch shall be placed in the normal position after installation.

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TABLE IV. Standard parts.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Lamp	Commercial identification F17T8/CW	6240-01-391-6027	
Lamplock	M16377/40-003		
Lampholder	UL 542	6250-00-284-0481	Commercial
Ballast	M16377/81-001		Integrated starter type
High-temperature ballast	M16377/81-002		High-temperature applications
High-temperature starter	FS-4	6250-00-179-2354	High-temperature applications
Startersocket	UL 496	6250-00-295-2738	High-temperature applications
Shock mount	M16377/41-008	6210-01-393-4302	0.067 steel, zinc plated
Window assembly, white diffusing	M16377/12-001	6210-00-295-2849	1/
Window assembly, clear black shielded	M16377/12-003	6210-00-725-9030	1/
Window assembly, clear and white prismatic	M16377/12-004	6210-00-814-5914	
Emergency ballast, wiring harnesses, and mating connector	M16377/77-001	6250-01-491-7039	Complete assembly: wiring harnesses and mating AMP #206043 connectors.
Charging indicator light		6220-01-415-7889	LED type
Switch, battery enable		5930-01-414-9240	DPST
Switch, battery test	Carling 2167-M-A-NC-73XCA4 or equal		Momentary toggle
Switch cover, rubber boot	MIL-DTL-5423/2		Neoprene
Filter, 17-watt T8 fluorescent lamp (Dark adaptation)	M16377/42-010	6210-01-313-8169	Red transparent
Filter, 17-watt T8 fluorescent lamp (1% transmission gray)	M16377/42-011		
Filter, 17-watt T8 fluorescent lamp (2.5% transmission gray)	M16377/42-012		
Filter, 17-watt T8 fluorescent lamp (Amber)	M16377/42-015		
Filter, 17-watt T8 fluorescent lamp (Broad band blue)	M16377/42-016		

NOTE:
1/ Window assemblies (diffusing and prismatic) are the same as T12 lamp fixtures (MIL-DTL-16377/12).

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TABLE V. First article tests and comparison inspection for complete fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Electromagnetic interference	3.6.14	4.8.16	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	[2/]
Voltage transient spike	3.6.18.2	4.8.23	[3/]
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/] Watertight
Shock	3.5.12	4.8.7	[4/]
Vibration	3.5.13	4.8.8	[4/]
Enclosure effectiveness	3.5.11	4.8.14.4	[4/] Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	[4/]
Insulation resistance	3.6.16	4.8.3	[4/]
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output: Distribution curves shall be documented during emergency lamp operation	3.7.1	4.8.6.1	[5/]
Emergency ballast performance: Operating time and discharge test	See MIL-DTL-16377/77, table II, battery discharge test	See MIL-DTL-16377/77, table II, battery discharge test	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ If less, then 1 KVA exception not applicable for input waveform total harmonic distortion.

3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes, (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300-1.

4/ Tests shall be performed in the order specified.

5/ Shall meet or exceed published distribution curves.

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TABLE VI. First article tests and comparison inspection for window assemblies. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/ Watertight
Light output	3.7.1	4.8.6.1	Candela distribution

NOTES:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.
2/ Tests shall be performed in the order specified.

TABLE VII. Complete fixture part numbers.

Military Part Number M16377/84-	Item Name	National Stock Number (NSN)
333.1T8EM	Complete fixture, Symbol 333.1T8EM	6210-01-591-9932
333.2T8EM	Complete fixture, Symbol 333.2T8EM	6210-01-595-4265
342.2T8EM	Complete fixture, Symbol 342.2T8EM	6210-01-595-4271
342.3T8EM	Complete fixture, Symbol 342.3T8EM	

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Custodians:
Army – CR4
Navy – SH
Air Force – 03

Review activities:
Navy – AS, CG
DLA – GS2

Civil agency:
GSA – FAS

Preparing activity:
Navy − SH
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